

House Price Prediction using Machine Learning

1. Objective

The objective of this project was to predict house prices using machine learning techniques. Different features of houses were analysed to understand their relationship with price and to develop models for accurate price prediction.

2. Dataset Information

The dataset used in this project contains information about houses and their characteristics. It consists of 545 rows and 13 columns. The target variable is **price**, while the remaining columns are used as input features for prediction.

Features Used

- Area
- Bedrooms
- Bathrooms
- Stories
- Main Road
- Guest Room
- Basement
- Hot Water Heating
- Air Conditioning
- Parking
- Preferred Area
- Furnishing Status

3. Data Preprocessing

The dataset was checked for missing values and duplicate records. No missing values or duplicate rows were found. Categorical columns were converted into numerical form using one-hot encoding so that they could be used for machine learning models. The cleaned dataset was then prepared for training and testing.

4. Models Used

Two machine learning models were implemented in this project:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio.

5. Model Evaluation

The performance of both models was evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

After comparing the results, the Linear Regression model performed better and provided more accurate predictions for this dataset.

6. Visualizations

Three charts were created for better understanding and analysis:

1. Histogram showing the distribution of house prices.
2. Correlation heatmap showing the relationship among different features.
3. Scatter plot comparing actual and predicted house prices.

7. Findings and Insights

The analysis showed that factors such as house area, number of bathrooms, air conditioning, and preferred area had a strong influence on house prices. Houses with larger areas and better facilities generally had higher prices. It was interesting to observe that location-related features and amenities also played an important role in determining house values. Among the two models, Linear Regression produced better results and gave more reliable predictions.

8. Recommendation

Based on the findings, real estate businesses should focus on properties with larger areas and better facilities, as these factors contribute significantly to higher house prices and increased customer demand.

9. Conclusion

This project demonstrated the application of machine learning techniques for house price prediction. By comparing Linear Regression and Random Forest Regressor, it was found that the Linear Regression model provided better predictive performance for this dataset. Overall, the project highlighted the importance of data analysis and machine learning in understanding the factors that influence house prices.